Can Global Commodity Markets Interoperate without a Central Gatekeeper?

A Federated, Vendor-Neutral Architecture and Testable Research Agenda

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Abstract

International commodity trade relies on many specialized markets, regulated payment providers, inspection agencies, carriers and document systems. Recent interoperability advances show that common interfaces can work, while fragmentation, asymmetric incentives and control over shared infrastructure remain unresolved. This paper asks whether reusable, commodity-agnostic trade capabilities can be federated among independently governed markets without introducing a new central exchange, proprietary currency or vertically integrated inspection authority. We develop a conceptual design framework with distinct market, standards, trust, services and governance layers; derive five architectural propositions and six falsifiable hypotheses; and specify an evaluation protocol comparing conventional bilateral integration, an incumbent-controlled platform and a neutral federation. Physical-origin evidence, identity, legal recognition, confidentiality, commercial incentives and cryptographic adaptability are treated as co-equal design constraints. India is examined as an economically relevant reference for heterogeneous markets and interoperable digital public infrastructure, not as an assumed sponsor or exclusive implementation geography. The intended scholarly contribution is a testable design and governance research agenda. It does not claim an implemented product, measured efficiencies, proof of market adoption or a novel cryptographic primitive.

Keywords: commodity markets; interoperable trade; platform governance; federated architecture; digital trade documents; crypto agility; digital public infrastructure; India; design science.

1. Introduction

What should be shared across the world's commodity markets, and what should remain under the control of the independent participants who trade in them? Grain, timber, mineral ores, energy products and metals differ in quality standards, delivery practices, market structures and regulatory obligations. Yet they often reuse common administrative functions: organizational identification, document authenticity, inspection attestations, event communication, payment coordination, claims handling and data reconciliation.

A closed platform can simplify coordination for members while making others dependent on a dominant operator. Direct bilateral integrations preserve autonomy but become expensive to maintain as the

network expands. A third possibility is a federated architecture in which reusable functions and rules connect independent systems without requiring a single firm to own trading relationships, determine prices or collect confidential deal data.

In October 2026, the ICC Digital Standards Initiative emphasized accelerating fragmentation and the need for interoperability, trust and identity [1]. The 2026 FIT Alliance survey reported that interoperability and fragmentation were the most frequently cited obstacles for electronic bills of lading (eBLs) [2]. Crucially, DCSA had already announced in June 2026 that five providers had adopted an eBL interoperability framework [3]. The research problem is therefore **not whether basic cross-platform exchange is imaginable**. It is whether a wider, multi-commodity, operationally sustainable and governance-neutral architecture can improve on fragmented and potentially concentrated models.

The paper addresses three questions: **RQ1:** Which capabilities can be reused across commodity verticals without erasing their legal and operational distinctions? **RQ2:** What institutional design might maintain interoperability while restraining infrastructure capture? **RQ3:** Under which testable conditions could such a federation outperform existing alternatives in cost, continuity and trust?

2. Scope, method and epistemic limits

This is an **analytical design-proposition study** informed by technical standards, selected market evidence and established research on modularity, institutional governance and information-systems design [4–6]. It follows the problem-and-artifact logic of design-science research but **does not claim to have built or evaluated an artifact** [6]. The contribution is a reference architecture and an evaluation protocol that independent investigators may implement and challenge.

Evidence is separated into three categories. **Established facts** include public standards, dated market initiatives and the closure of particular platforms. **Design propositions** specify what an independent implementation might attempt. **Unverified hypotheses** predict outcomes that can be rejected by observation. No selection of cases, interviews, transaction datasets, code repository, security audit or operational pilot was undertaken for this paper. A standards and literature review was selective rather than systematic; a later systematic review may challenge the novelty or utility of the proposed combination.

The institutional role of Swiss Family Office (Novel Technology) Club (SFOC) is limited to independent concept publication, professional dialogue and networking. SFOC does **not** develop, finance, own, operate, validate, govern or control any future trading infrastructure. Independent actors alone would determine whether to undertake technical work, funding, regulatory qualification and commercial adoption.

3. Prior work and the actual gap

3.1 Operational interoperability exists

DCSA's June 2026 announcement identifies five eBL providers using a shared interoperability annex [3]. The framework couples technical transfer with legal arrangements and control tracking, making it particularly instructive: document portability is a socio-technical issue, not merely an API issue. ICC's Key Trade Documents and Data Elements initiative also addresses common semantics across trade documents [7]. These efforts are prior work to complement, not competitors to overwrite.

3.2 Distinct commercial verticals also exist

Covantis addresses post-trade workflows in agricultural commodities, VAKT operates in physical commodity post-trade workflows, and Komgo supports digital trade finance processes [8–10]. Their different scopes demonstrate demand for digital services in individual parts of trade. They do **not** demonstrate that demand exists for one universal Engine; the architecture must prove additional value such as cheaper independent integration, multi-vertical portability or superior exit resilience.

3.3 Platform failure and institutional incentives

Maersk and IBM announced the discontinuation of TradeLens in November 2022, citing an inability to reach necessary global industry collaboration and commercial viability despite a technically viable platform [11]. The record does not prove that incumbent ownership *caused* failure. It does motivate investigation into incentive compatibility, credible neutrality, data custody and the risks of a dominant sponsor. Work on collective-action institutions [4] and modular design [5] suggests that transparent rules and separable components could be relevant, but applicability to global commodity infrastructure must be tested rather than presumed.

3.4 Legal and identity building blocks

UNCITRAL's Model Law on Electronic Transferable Records (MLETR) provides technology-neutral principles for electronic transferable records, including reliable exclusive control [12]; the rules have to be implemented or otherwise recognized in relevant national legal systems. W3C's Verifiable Credentials Data Model v2.0 supplies a standardized framework for verifiable claims [13]. The Global Legal Entity Identifier Foundation's vLEI provides a mechanism for verifying organization and role assertions [14]. These are optional, complementary building blocks, not evidence that signatures alone determine ownership of physical goods.

Identified research gap: Can a reusable combination of cross-vertical semantics, legally cognizant evidence exchange, independent governance and genuine provider substitution create measurable benefits beyond existing vertical platforms and interoperable document networks? No positive answer is assumed.

4. Design requirements and falsifiable boundaries

R1 — Non-exclusivity. No mandatory central exchange, single broker, central price setter or sole trading counterparty. Existing bilateral trade and independent marketplaces remain possible.

R2 — Institutional continuity. Existing banks, inspection bodies, certifiers, insurers and logistics firms keep their independent functions and responsibilities. A digital signature confirms provenance and integrity of a claim, not physical correctness.

R3 — Payment restraint. The architecture does not issue a proprietary currency, cryptocurrency or payment instrument. Execution remains through lawful financial systems using recognized sovereign currencies; any digital form requires separate legal assessment.

R4 — Modularity and exit. Interfaces must be documented, versioned and testable. Adapter switching, data export, trust-anchor replacement and continuity of active transactions must be examined in an actual exit experiment.

R5 — Confidentiality. Trading prices, customers, quantities and strategic information should not become shared public data merely because transaction evidence is verifiable.

R6 — Legal plurality. Commodity rules, transfer-of-title requirements, trade documents and dispute procedures remain sensitive to instrument and jurisdiction [12]. Technical finality and legal finality are not identical.

R7 — Cryptographic agility. NIST's 2026 updated guidance provides a reference for replacing cryptographic mechanisms while maintaining operations [15]. NIST's post-quantum standards FIPS 203, 204 and 205 provide candidates for evaluation, not proof of end-to-end quantum resistance [16].

R8 — Anti-capture. No initial technology partner, industry sponsor, investor or public body should be able to unilaterally change network rules, compel access to all confidential transactions or make supplier exit impossible.

5. Proposed reference architecture

A future implementation might separate five logical planes; these are **functions, not a requirement for five databases or one shared blockchain**.

Plane	Core responsibilities	Ownership and trust boundary
Market/vertical	Product taxonomy, grades, quality conventions, workflows, delivery and dispute rules	Autonomous commodity communities and trading participants
Interoperability	Versioned data schemas, event contracts, adapter catalogue, document portability	Open technical standards, multiple implementers
Identity and evidence	Organizational authority, signed inspection results, attestation lifecycle, revocation	Qualified external issuers and authorized counterparties
Service integration	Payment status, financial institutions, inspection, logistics, insurance, customs	Existing regulated or professional actors
Coordination and governance	Protocol changes, interoperability conformance, continuity, security and appeals	Accountable and independently represented constituencies

The design should avoid a compulsory central global ledger. Independent parties might use signed messages, registries, federated event systems, restricted ledgers or other technologies according to the function. Only evidence that truly needs shared consensus should be placed in a jointly maintained state. A blockchain, if chosen, must justify its incremental benefit against less complex designs.

The suggested architecture separates **three distinct decentralization claims**. Distributed computation means independent organizations may process or host services. Decentralized governance means no single actor can rewrite the shared rules. Consensus refers to agreement on a limited shared state, when needed. One property does not imply the others.

6. Evidence, title, privacy and governance

6.1 Digital trust does not repair physical data

A warehouse intake report can be digitally authentic yet factually wrong. Therefore, a verifiable record should identify source, issuer, authorized role, commodity lot, measurement procedure, timestamp, qualifications, uncertainty and corrective events. Sampling, calibration and independent dispute review remain physical and institutional work. First-mile data quality is an essential condition of the architecture.

6.2 Legal controls must precede automation

For transferable records, legal standing depends on the relevant legislation and contractual context [12]. Permission to inspect, ownership of goods, authority to release cargo and legal control of an eBL may belong to different actors. Workflow automation must reflect those distinctions. Banking integration must not be conflated with guarantees of atomic payment or elimination of counterparty risk.

6.3 Minimal governance design

An independent implementation could test a governance model representing trading users, service providers, operators and independent technical/legal expertise. It should specify eligibility, constituency balance, proposal handling, documented conflict recusals, supermajorities for constitutional protocol changes, appeals, security incident powers with expiry, funded maintenance and supplier-removal processes. Token ownership alone should not determine voting power. Collective-action theory explains why neither informal coordination nor a nominally neutral foundation guarantees fair outcomes [4].

6.4 Provider-switch as the decisive operational test

A network is not meaningfully open merely because its APIs are public. An end-to-end test should remove one participating identity provider, inspection connector or core technology supplier, transfer active and archived records and verify that independent parties can continue without new compulsory gatekeepers. Residual dependencies should be reported as failures or qualifications, not ignored.

7. Commercial adoption and regional application

Interoperability creates value only if each participant expects benefits that exceed adoption and switching costs. Possible benefits include fewer duplicate interfaces, faster document reconciliation, fewer provenance disputes and easier access for previously excluded smaller counterparties. Possible costs include conformance testing, governance contributions, privacy risks, liability negotiations and support. A demonstrator may work while commercially viable adoption fails, as the TradeLens case warns [11].

A lean, corridor-first implementation by independent parties is a plausible alternative to attempting immediate world-scale replacement of established networks. It is **a strategic hypothesis, not a finding that a small project will necessarily succeed**. A selected corridor must have consented participants, legally feasible workflows, quantifiable baseline friction and a documented right to withdraw.

India as a substantive reference case. India combines large agricultural trading networks with experience in interoperable digital public infrastructure. The Government of India reported 1,656 e-NAM mandis across 23 states and four union territories by March 2026 [17]; Trade Connect supplies a separate

example of export-service digitization [18]. These are not GCTE implementations or proof that decentralized international commodity settlement works. Rather, they make India an especially informative location for studying coexistence among legacy markets, standard interfaces, digital identity and cross-border trade needs. India-linked corridors should compete under objective criteria with proposals from Africa, Latin America, Europe and elsewhere; selection is commercially grounded and politically neutral.

The Asian Development Bank's 2025 survey estimates a USD 2.5 trillion global trade finance gap [19]. The African Union's AfCFTA digital-trade protocol advocates interoperable standards [20]. The September 2026 BRICS New Delhi Declaration identifies trade-document digitization and access to finance as policy interests of its signatories [21]. These sources describe market context; they imply neither SFOC alignment with a political grouping nor endorsement, funding or future participation by any institution.

8. Testable hypotheses

H1 — Workflow interoperability. A federated implementation reduces duplicate interfaces or reconciliation actions for a matched multi-party workflow relative to bilateral point-to-point integrations.

H2 — Provider substitutability. A standards-compliant provider can be replaced, preserving authorized access to active and archived trade evidence, within a predeclared recovery threshold without centralized intervention.

H3 — Commercial confidentiality. Independent cross-platform verification can be performed without revealing protected prices, customer relationships or full trade histories to unrelated network operators.

H4 — Governance resistance. Under simulated concentrated funding and voting pressure, the governance protocol prevents unilateral changes to market access, access policies or fees.

H5 — Cryptographic adaptability. Defined document-verification workflows remain valid through controlled algorithm and trust-anchor migration, subject to measurable overhead and legal constraints.

H6 — Independent adoption. At least one corridor demonstrates net participant value and a viable independent multi-party operating model without imposing a mandatory central venue, new money or exclusive inspection provider.

Failure of an experiment should invalidate or qualify its corresponding claim; targets should not be revised after observing results without disclosure.

9. Evaluation protocol for independent researchers

A research team should **pre-register** test scenarios and success criteria before building its demonstrator. The appropriate comparison is between: **B0** existing paper/manual or bilateral integration; **B1** a conventional centrally governed shared platform; and **B2** an independently governed standards-based federation. The comparison must use equivalent legal, security and transaction requirements, otherwise lower costs may simply reflect missing safeguards.

Metric	Measurement approach	Failure or limitation to report
Integration burden	Number of bespoke connections; developer-hours for equivalent workflow	Hidden proprietary dependencies

Metric	Measurement approach	Failure or limitation to report
Process efficiency	Median elapsed document and reconciliation times	Improved speed but more errors
Source-trust quality	Traceability, issuer revocation and corrected-record handling	Valid signature over false evidence
Participant autonomy	Concentration of voting/control rights; permission changes	Effective veto by one sponsor
Confidentiality	Adversarial access and metadata-leakage tests	Counterparty or price inference
Exit resilience	Recovery time, evidence retention and provider replacement	Provider exit breaks open transactions
Cryptographic transition	Signature/key rotation costs, old-record verification, interop failures	Archive evidence no longer verifiable
Overall economics	Participant-level full costs and repeatable funding model	Savings for some funded by uncompensated others

The study should distinguish laboratory tests from live legal and financial transactions. No live settlement should occur without appropriately licensed institutions and jurisdiction-specific legal assessment. For a live pilot, investigators should publish a threat model, conformance profile, privacy treatment, independent-party count, adversarial test log, sample definition, error bars and negative results. A 12–18 month sequence sometimes proposed for exploration is not a prediction or SFOC delivery commitment.

10. Risks, rival explanations and falsification

Several rival explanations could defeat the proposed approach. Existing DCSA and vertical platforms may already be sufficient; adding an Engine could increase complexity without benefit. Governance that disperses voting could become slow, underfunded or captured through informal financing. A neutral protocol might have no sustainable maintenance budget. Cross-border legal recognition may prevent a uniform transfer workflow. Compliant verifiable credentials could still attest to fraudulent physical goods. Public metadata or inter-operator logs could expose commercially sensitive information. Post-quantum migration may produce performance or interoperability costs greater than its near-term benefit.

The strongest falsification condition is straightforward: if a standards-based federation cannot show **incremental value over existing interoperable solutions without sacrificing independence, legal certainty, confidentiality or sustainable economics**, then the additional architectural layer is not justified. Scientific credibility requires publishing that outcome as readily as a positive one.

11. Discussion and future work

The conceptual innovation is not a new cryptographic algorithm, legal standard, payment asset or commodity exchange. It is the **joint design question**: whether modularity, institutionally separate services, credible governance neutrality and empirically demonstrated supplier exit can support federation across distinct commodity markets. The distinction between a reusable capability and a globally controlled application is central. The architecture needs stronger formal threat modelling,

comparative governance research, jurisdiction-by-jurisdiction legal mapping and independent transaction-data studies before its practical superiority can be claimed.

Future work can begin narrowly: one trade-document workflow, two independent implementers, verified organizational and inspection attestations, explicit legal boundaries, a privacy-preserving proof of exchange, an exit test and an external reviewer. The next vertical should not be chosen until the first demonstrates measurable value and operational continuity. SFOC's contribution remains conceptual publication and neutral professional exchange; independent institutions, if interested, would separately make every implementation and financing decision.

12. Conclusion

Global commodity trading would not become more open simply by substituting a new blockchain provider for an existing centralized intermediary. The relevant research question is whether shared services can be **portable, verifiable, economically attractive and resistant to institutional capture** while leaving commodity-specific commerce under the control of its legitimate participants. Existing 2026 evidence warrants serious investigation of this question but does not answer it. This final conceptual article provides a falsifiable design framework and measurement protocol; empirical proof, if any, remains for independent future work.

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